# RESOURCE LETTER

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## Resource Letter PB-1: The Physics of the Brain

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This Resource Letter provides an overview of the literature relating physicists' past, present, and future contributions to our understanding of the brain. It is intended as a starting point for upper-level physics students and physicists who are interested in learning about and/or investigating the brain. Topics include the biophysics of individual neurons, models of learning and memory, structure and dynamics of brain networks, brain-inspired computing hardware, and the possible role that quantum mechanics may play in brain function. © *2018 American Association of Physics Teachers.*
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### I. INTRODUCTION

We live in a golden age for studying the brain. Month by month, fundamental discoveries accrue, shedding light on the mysteries of cognition, computation, consciousness, and what it means to be human. Despite the speed of this progress, contemporary neuroscience is at perhaps the same stage of development as mid-nineteenth century physics, resting on a solid foundation of core knowledge, but waiting for explosive paradigm shifts to recast how these facts and figures fit together. Physicists have played, and will continue to play, a crucial role in this endeavor. We offer not only useful theoretical and experimental tools, but also a way of thinking that is uniquely suited to tackling many difficult problems related to the brain.

With an estimated $86 \times 10^9$ neurons and more than $\sim 10^{14}$ synaptic connections, the human brain is enormously complex. Investigating the biological minutiae of this system has undoubtedly furthered our understanding of it, but stripping away biological detail in order to focus on core mechanisms of brain function has also proven effective. In 1952, for example, biophysicists Alan Hodgkin and Andrew Huxley distilled the electrical properties of the lipid bilayer, myriad ion channels, and various ion pumps composing a neuronal cell membrane to a simple electrical circuit.[12] The Hodgkin-Huxley model was described by an elegant system of four differential equations. This powerful paradigm is still used in its original form to understand neuronal dynamics. In the 1970's, evidence mounted that long-term memories are stored through modification of the strengths of neuronal connections, but no one understood how this could underpin the ability of a network to reliably store multiple memories. Then in 1982, John Hopfield drew inspiration from the Ising model to formulate an elegant theory of "auto-associative memory." He showed how a simple rule for modifying connection strengths based on neuronal activity enables large-scale memory storage.[16]

The theories of Hodgkin, Huxley, and Hopfield are excellent examples of the kinds of unifying frameworks we need if we are to attain ultimate understanding of the brain. The recent work of Eric Jonas and Konrad Kording,[4] investigating the function of a primitive microprocessor using analogs of modern neuroscientific techniques, highlights our current shortage of such unifying principles. They found that although such techniques *described* the microprocessor at various scales (such as characterizing the operation of a single transistor, or exhaustively tabulating the pattern of connections between transistors), it was impossible to construct a "unified theory" of the microprocessor's operation without a solid theoretical foundation. Therefore, even as techniques are developed to obtain a more comprehensive view of brain activity (an effort to which physicists are making significant contributions—see Refs. 110–113), it is important to recognize that we need new paradigms for understanding the resulting deluge of data. If the history of neuroscience is any guide, physicists should play an important role in developing such unifying frameworks going forward.

The father of neuroscience, Ramòn y Cajal, developed the foundational framework for our current understanding of brain function over one hundred years ago. The results of his pioneering experiments convinced him that: (1) the brain is composed of individual neurons (as opposed to one super-network of biological processes), (2) neurons form connections, or *synapses*, with one another, (3) synapses allow for *one-way* transmission of signals, and (4) these synapses form neuronal circuits to process information.[2] Later, it was discovered that neurons nonlinearly integrate input from the synaptic connections they receive. If the net input is

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sufficient to drive the neuron's membrane potential above some threshold (usually 10 to 20 mV above the cell's resting membrane potential, which is often around −70 mV), the neuron undergoes an action potential. This results in electrochemical signals being transmitted to all the post-synaptic neurons it sends connections to. Importantly, a neuron may be either *inhibitory*, in which case its signal drives a post-synaptic neuron's membrane potential away from firing threshold, or *excitatory*, in which case its signal makes a post-synaptic neuron more likely to fire. From an information processing perspective, the brain may be viewed as a network of non-linear summation units.

There are, of course, layers upon layers of complexity beyond this superficial description. While an exhaustive review of "the physics of the brain" would span many volumes, this Resource Letter attempts to highlight the most important insights contributed by physicists to our current understanding of brain function.

*Note*: This paper often uses the adjective "neuronal" rather than "neural," since "neural networks" are too easily confused with algorithms developed by computer scientists, rather than the biological brain networks which are this Resource Letter's focus.

## II. BASIC RESOURCES

### A. Journals

*Journal of Computational Neuroscience*
*Journal of Neurophysiology*
*Journal of Neuroscience*
*Nature*
*Nature Neuroscience*
*Nature Physics*
*Neural Computation*
*Physical Review Letters*
*Physical Review E*
*PLoS Computational Biology*
*Proceedings of the National Academy of Sciences*
*Science*

### B. Workshops/Courses

*Woods Hole Marine Biological Laboratory: Methods in Computational Neuroscience*. This annual four-week summer course introduces fundamental concepts in computational neuroscience from both experimental and theoretical perspectives. Students acquire facility with core computational tools, and, in the last week, they apply their skills to a particular project. http://www.mbl.edu/mcn/

*Cold Spring Harbor Laboratory neuroscience courses*. Cold Spring Harbor hosts a plethora of neuroscience-related summer courses for those interested in entering the field. Past courses have included Neural Data Science and Computational Neuroscience of Vision. http://meetings.cshl.edu/courseshome.aspx

### C. Getting started

The following resources are recommended as a preliminary introduction to neuroscience for physics students and physicists. Together they describe the core biological principles of brain function, a brief history of how physicists migrated to neuroscience, fundamental results obtained thus far, and the need to develop robust theories of brain function.

1. "Theoretical neuroscience rising," L. F. Abbott, Neuron **60**(3), 489–495 (2008). Gives a historical and conceptual overview of the field of theoretical neuroscience, from the perspective of a theoretical physicist. (E)
2. **In Search of Memory: The Emergence of a New Science of Mind**, E. R. Kandel (W.W. Norton & Company, New York, 2007). Gives an overview of essential biological details related to the brain. (E)
3. **Theoretical Neuroscience**, P. Dayan and L. F. Abbott (MIT Press, Cambridge, MA, 2001). An introduction to the field of theoretical neuroscience, co-authored by a physicist. While a bit light on the biophysics of individual neurons, it presents an excellent introduction to neuronal encoding and decoding, as well as plasticity and learning. (E)
4. "Could a neuroscientist understand a microprocessor?," E. Jonas and K. P. Kording, PLoS Comput. Biol. **13**(1), e1005268 (2017). Applies contemporary techniques from neuroscience to attempt to understand how a primitive microprocessor works, suggesting the need for improved theories of brain function. (I)
5. **The Handbook of Brain Theory and Neural Networks**, M. A. Arbib (editor) (MIT Press, Cambridge, MA, 2003). Offers 287 self-contained articles covering a wide range of topics, including models of individual neurons, language, sensory and motor systems, and artificial neural networks. (I)
6. **Biophysics: Searching for Principles**, W. Bialek (Princeton U.P., Princeton, 2012). Graduate-level text that explores fundamental principles underpinning the operation of biological and neural systems, with an emphasis on information theoretic techniques. (A)

## III. MODELING INDIVIDUAL NEURONAL DYNAMICS

Perhaps physicists' most fundamental contribution to our understanding of the brain has been the elucidation of the biophysical processes underpinning action potential generation. Before this discovery, individual neurons were often modeled as RC circuits with a hard-coded voltage threshold that, when breached, resulted in (1) spiking and (2) resetting of the voltage to a baseline value (see Figs. 1(a) and 1(b)). In this model, the so-called leaky integrate-and-fire model of the neuron, a capacitor is used to model the cell membrane, which at rest collects positive ions on its external surface and negative ions on its internal surface. It is the potential difference across the capacitor that constitutes the "membrane potential." The resistor represents ion channels that allow current to flow through the cell membrane, while the battery enforces the empirical observation that the electric potential inside a resting neuron is typically 70 mV below the potential outside. (This gives a "resting membrane potential" of −70 mV, since the extracellular medium is always assumed to be the electrical reference.) Finally, the current source represents current experimentally injected by an electrode. This "leaky integrate-and-fire" model, however, does not attempt to describe the dynamics of the action potential itself.

Such a model had to wait for the pioneering experiments of the biophysicists Alan Hodgkin and Andrew Huxley. Investigating the squid giant axon in the early 1950s,[12] Hodgkin and Huxley showed that action potentials are the result of a carefully choreographed dance between depolarizing sodium currents and repolarizing potassium currents (see

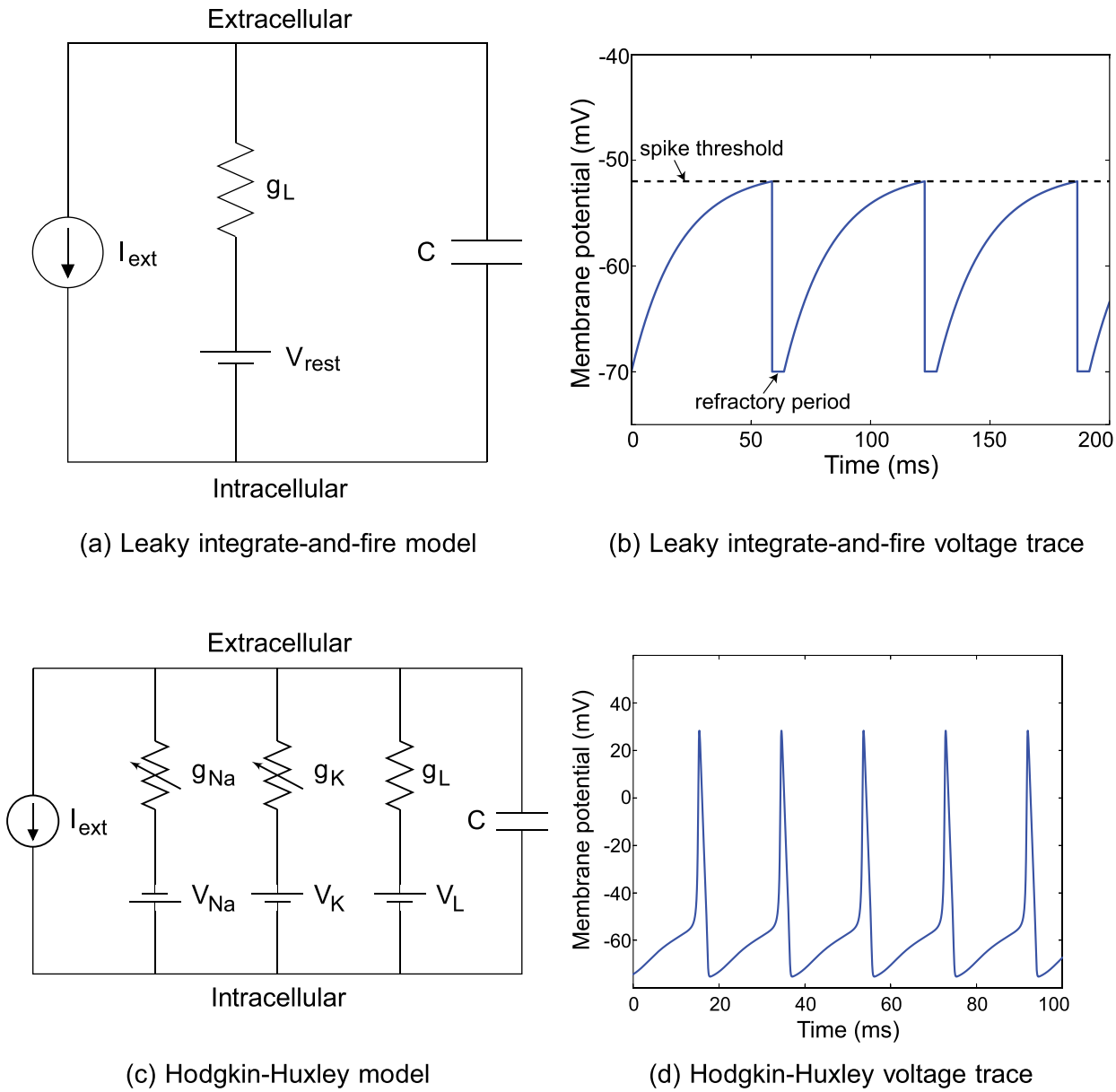



Fig. 1. Two models of neuronal dynamics. (a) The leaky integrate-and-fire neuron model, which uses a capacitor to model the cell membrane, the potential difference across the capacitor to reflect the "membrane potential," a resistor to represent ion channels, and a battery to enforce the resting membrane potential. (b) The leaky integrate-and-fire neuron exhibits repetitive spiking when current is injected. However, the spike itself is simply *assumed* to occur whenever the membrane potential breaches some threshold. The membrane potential is then re-set to its resting value, where it is held for several milliseconds to model the "refractory period," during which the neuron is incapable of spiking, no matter how strongly it is stimulated. (c) The Hodgkin-Huxley model adds sodium ($Na^+$) and potassium ($K^+$) ionic currents whose associated conductances ($g_{Na}$ and $g_K$) are nonlinear functions of the membrane potential. (d) The Hodgkin-Huxley model accounts for neuronal dynamics both below threshold *and* during the action potential itself. It also explains *why* a threshold for spike generation exists, as well as why neurons exhibit refractory periods. (Adapted from Ref. 42.)

Figs. 1(c) and 1(d)), described by an elegant system of four ordinary differential equations. The Hodgkin-Huxley model, therefore, adds two crucial currents to the integrate-and-fire model: the flow of sodium ($Na^+$) and potassium ($K^+$) ions. Concentration gradients result in sodium ions naturally flowing into the cell (resulting in depolarization) and potassium ions flowing out of the cell (causing repolarization). Only when the membrane potential breaches a threshold do the sodium ion channels appreciably open, which is modeled as a significant increase in the voltage-dependent sodium conductance $g_{Na}$. This causes a rapid rise in the membrane potential, which is short-lived because $g_{Na}$ returns to 0 just as $g_K$ is activated, resulting in the neuron's return to rest. For their efforts, Hodgkin and Huxley were awarded the 1963 Nobel Prize in Physiology or Medicine,[13] and their framework for understanding neuronal dynamics has endured to this day.


7. **Biophysics of Computation**, C. Koch (Oxford U.P., New York, NY, 1999). A good introduction to modeling the biophysical processes of individual neurons, with an emphasis on how such processes contribute to cellular computation. The section that applies linear cable theory to explain dendritic function is particularly insightful. (E)
8. **Foundations of Cellular Neurophysiology**, D. Johnston and S. M. Wu (MIT Press, Cambridge, MA, 1995). An

excellent resource for learning about the biophysics of ion movement and the modeling of neuronal electrical properties. The first one-third of the book lays a foundation for understanding the Hodgkin-Huxley neuron model. (I)

9. **Dynamical Systems in Neuroscience**, E. M. Izhikevich (MIT Press, Cambridge, MA, 2007). Presents more exotic neuronal models (such as the FitzHugh-Nagumo, Izhikevich, and Hindmarsh-Rose models), many of which are simplifications of the Hodgkin-Huxley model. Emphasizes the bifurcation structure involved in neurons' transition to firing. (I)
10. "On numerical simulations of integrate-and-fire neural networks," D. Hansel, G. Mato, C. Meunier, and L. Neltner, Neural Comput. **10**(2), 467–483 (1998). Relates best practices for numerically simulating integrate-and-fire neurons. (A)
11. "Stochastic integrate and fire models: A review on mathematical methods and their applications," L. Sacerdote and M. T. Giraudo, in **Stochastic Biomathematical Models**, edited by M. Bachar, J. J. Batzel, and S. Ditlevsen (Springer, Berlin Heidelberg, 2013), pp. 99–148. Reviews methods for analyzing and simulating stochastic integrate-and-fire neuron models. (A)
12. "A quantitative description of membrane current and its application to conduction and excitation in nerve," A. L. Hodgkin and A. F. Huxley, J. Physiol. **117**(4), 500–544 (1952). The original paper presenting the Hodgkin-Huxley model of neuronal dynamics. (A)
13. "The quantitative analysis of excitation and conduction in nerve," A. F. Huxley, Les Prix Nobel, 242–260 (1963). (I)

## IV. MODELS OF LEARNING AND MEMORY

Aside from the Hodgkin-Huxley model of neuronal dynamics, the most fruitful cross-pollination between physics and neuroscience has been John Hopfield's model of auto-associative memory. Inspired by the Ising model (which models ferromagnetism by assuming that the spin state of an atom is influenced by the spin states of its neighbors), it assumes a fully connected network of neurons that learns to recall specified activation patterns by adjusting connection strengths according to co-variation of neural activity. That is, if two neurons are simultaneously active or inactive within a particular pattern, then the connection strength between them increases, while if one is active and the other inactive, the connection strength decreases. The resulting pattern of connectivity produces local minima in an abstract energy landscape, to which an initial state resembling one of the learned states will be attracted. Therefore, this form of memory is said to be "content-addressable," since the recall of a memory depends on the content of an initial cue being similar to that of a previously learned state. For example, this models the everyday experience of hearing part of a song and remembering what comes next.

Hopfield's model gave theoretical teeth to the idea, proposed by Donald Hebb over three decades earlier,[15] that coupling between neurons is modified by neuronal activity, and that such "activity-dependent plasticity" is the key to forming long-term memories. Neuroscientists have since proposed that the CA3 region of the hippocampus may effectively function as a Hopfield network,[20] though experimental limitations make this hypothesis difficult to investigate. Hopfield's work revived research into artificial neural networks after interest had waned in the 1970s, thereby laying the foundation for the explosion in machine learning research we see today, with many synaptic learning rules and neuron models exhibiting features of memory and learning.

14. "Hopfield network," J. J. Hopfield, http://www.scholarpedia.org/article/Hopfield_network **2**(5), 1977 (2007). Overview of the Hopfield model of content addressable memory. (E)
15. **The Organization of Behavior: A Neuropsychological Theory**, D. O. Hebb (Wiley and Sons, New York, NY, 1949). (E)
16. **Tutorial on neural systems modeling**, T. J. Anastasio (Sinauer Associates, Incorporated, 2009). An introduction to Hopfield networks and many other brain-inspired machine-learning algorithms, suitable for lower-level undergraduate students. (E)
17. "An interactive simulation program for exploring computational models of auto-associative memory," C. G. Fink, J. Undergraduate Neurosci. Edu. **16**(1), 1–5 (2017). Presents a simulation program for building intuition with Hopfield networks. (E)
18. "Brain cells for grandmother," R. Q. Quiroga, I. Fried, and C. Koch, Sci. Am. **308**(2), 30–35 (2013). Gives a nice overview of recent experiments investigating network activity associated with memory recall. (E)
19. "Neural networks and physical systems with emergent collective computational abilities," J. J. Hopfield, Proc. Natl. Acad. Sci. **79**(8), 2554–2558 (1982). Original paper presenting the Hopfield model of learning and memory. (I)
20. "Memory retrieval time and memory capacity of the CA3 network: Role of gamma frequency oscillations," L. de Almeida, M. Idiart, and J. E. Lisman, Learning Memory **14**(11), 795–806 (2007). Explores the possibility that the CA3 region of the hippocampus may function as a Hopfield network. (I)
21. **Introduction to the Theory of Neural Computation**, A. Krogh, J. Hertz, and R. G. Palmer (Addison-Wesley, Redwood City, CA, 1991). Describes various neural network learning rules and algorithms that succeeded the Hopfield model, and reviews the contributions of statistical physics to the development of artificial neural networks. (I)
22. "Models of synaptic plasticity," H. Z. Shouval, http://www.scholarpedia.org/article/Models_of_synaptic_plasticity **2**(7), 1605 (2007). Overview of various approaches for more detailed modeling (both biophysical and phenomenological) of variable synaptic strength. (I)
23. "Spin-glass models of neural networks," D. J. Amit, H. Gutfreund, and H. Sompolinsky, Phys. Rev. A **32**(2), 1007–1018 (1985). Applies methods for analyzing spin glasses to Hopfield networks, deriving the critical temperature below which stored memories may be corrupted when recalled. (A)
24. "A learning algorithm for Boltzmann machines," D. H. Ackley, G. E. Hinton, and T. J. Sejnowski, Cognit. Sci. **9**(1), 147–169 (1985). Introduces the idea of the Boltzmann machine, which is similar to a Hopfield network except that its dynamics are stochastic rather than deterministic. (A)

**25.** "Weak pairwise correlations imply strongly correlated network states in a neural population," E. Schneidman, M. J. Berry, R. Segev, and W. Bialek, Nature **440**(7087), 1007–1012 (2006). Presents a "maximum entropy" generalized Ising model, which allows for specification of pairwise correlations between non-neighboring neurons and shows that such pairwise correlations alone are sufficient to explain collective activity in neuronal networks. (A)

## V. ANALYZING BRAIN STRUCTURE USING NETWORK THEORY

Beginning with Watts and Strogatz's study of "small world networks"[28] and Barabasi and Albert's investigation of "scale-free networks"[29] nearly twenty years ago, network theory has enjoyed an explosion of research interest. Combining mathematical graph theory with tools from statistical physics, network theory has found applications in biochemistry, ecology, epidemiology, genetics, sociology, and—of course—the study of the brain. At the cellular level, the brain can be thought of as a weighted, directed network, since synapses impose unidirectional information flow and their coupling strengths are variable. Unfortunately, current technology severely limits our ability to determine neuronal connectivity at this scale. High-resolution data are obtained by slicing brain tissue into sections tens of nanometers thick, using electron microscopy to image each section, and then stitching the images together using a combination of crowd-sourcing and machine learning. This laborious process realistically only yields the connectivity of $\sim$1000 neurons, however. Over thirty years ago, this method was applied to the roundworm *C. elegans*, still the only organism whose neuron-by-neuron "connectome" (of 302 cells) is completely known.

Brain connectivity can also be investigated at a much larger scale, considering tracts of axonal fibers connecting entire brain regions. Such fibers may be detected in the human brain using diffusion spectrum imaging (see Sec. XII), a technology employed by the Human Connectome Project to construct a "meso-scale" map of the human brain.[35] There is one primary drawback to diffusion spectrum imaging, however: information regarding directionality of connections is lost. Such information is preserved in tract-tracing, in which tracer molecules are injected in one part of the brain and light microscopy is used to detect their axonal diffusion. This technique has been applied to map the mesoscopic connectivity of the macaque cortex[36] and the entire mouse brain.[37] In addition, functional magnetic resonance imaging (fMRI, see Sec. XII) is often used to deduce "functional connectivity," which considers two brain regions to be "connected" if their mutual activity is significantly correlated, regardless of whether or not a structural connection actually exists between them.

All scales of analysis have identified "small world" properties in neuronal networks, in which most connections form clusters between nearby nodes, but some span much larger distances to connect disparate modules. Only a small fraction of such long-range connections is necessary to make the average shortest path between any two nodes nearly as small as in a randomly connected network. Many neuronal networks have also been shown to be "scale-free," i.e., the distribution of the number of nodal connections follows a power law, with the distribution's "fat tail" representing highly connected hub nodes. While the functional significance of these properties is still being worked out, it has been suggested that brain networks perform a constrained optimization, maximizing efficiency of information transfer while minimizing wiring costs. Long-range connections facilitated by hub cells aid in rapid information transmission, and the preponderance of short-range connections helps keep wiring costs low.[40] Another (perhaps complementary) framework views the brain's connectivity structure as balancing integration of information (via hub cells) with segregation of processing (reflected in locally connected modules).[33] Beyond such general speculation, however, we still understand very little about how neuronal structure underpins brain function.

**26. Networks of the Brain**, O. Sporns (MIT Press, Cambridge, MA, 2016). Overview of network theory techniques as applied to brain networks, plus descriptions of fundamental results from investigations into brain structure. (E)

**27. Connectome: How the Brain's Wiring Makes Us Who We Are**, S. Seung (Houghton Mifflin Harcourt, Boston, MA, 2012). Popular treatment of Seung's claim that individual identity is determined by the structural connectivity between neurons in one's brain. Includes descriptions of contemporary methods for elucidating neuronal connectivity. (E)

**28.** "Collective dynamics of 'small-world' networks," D. J. Watts and S. H. Strogatz, Nature **393**(6684), 440–442 (1998). This paper is a cornerstone of network theory, mathematically defining what it means for a network to exhibit the "small world" property and exploring the structural properties of the *C. elegans* neuronal network. (I)

**29.** "Emergence of scaling in random networks," A. L. Barabási and R. Albert, Science **286**(5439), 509–512 (1999). The paper is the other cornerstone of network theory, showing how power-law connectivity distributions naturally emerge from a growth process in which new nodes preferentially attach to existing nodes that are already highly connected. (I)

**30.** "Complex brain networks: Graph theoretical analysis of structural and functional systems," E. Bullmore and O. Sporns, Nat. Rev. Neurosci. **10**(3), 186–198 (2009). Reviews key methods for analyzing the structure of brain networks, as well as for processing the raw data needed to define connectivity in the first place. (I)

**31.** "Small-world brain networks," D. S. Bassett and E. D. Bullmore, Neuroscientist **12**(6), 512–523 (2006). Reviews evidence for small-world structure in the cortex, based on tract-tracing studies in the macaque and cat brains. (I)

**32.** "The economy of brain network organization," E. Bullmore and O. Sporns, Nat. Rev. Neurosci. **13**(5), 336–349 (2012). Reviews evidence that brain network topology results from a trade-off between optimizing information transmission while minimizing wiring and energy costs. (I)

**33.** "Rethinking segregation and integration: Contributions of whole-brain modelling," G. Deco, G. Tononi, M. Boly, and M. L. Kringelbach, Nat. Rev. Neurosci. **16**(7), 430–439 (2015). Reviews how an information theoretic approach is used to quantify integration and segregation of information processing in the brain. (I)

**34.** "Connectomic reconstruction of the inner plexiform layer in the mouse retina," M. Helmstaedter, K. L. Briggman, S. C. Turaga, V. Jain, H. S. Seung, and W. Denk, Nature **500**(7461), 168–174 (2013). Uses a combination of crowd-sourcing and machine learning to reconstruct electron microscopy sections of the mouse retina, ultimately elucidating the detailed connectivity structure between 950 neurons. (A)

**35.** "The Human Connectome Project's neuroimaging approach," M. F. Glasser *et al.*, Nat. Neurosci. **19**(9), 1175–1187 (2016). Outlines the methodology for gathering, processing, and disseminating the data necessary to construct meso-scale structural and functional human connectomes. (A)

**36.** "CoCoMac 2.0 and the future of tract-tracing databases," R. Bakker, T. Wachtler, and M. Diesmann, Front. Neuroinf. **6**(30), 1–6 (2012). Describes a database for aggregating results from dozens of experiments to construct a macroscopic connectome of the macaque brain. (A)

**37.** "A mesoscale connectome of the mouse brain," S. W. Oh *et al.*, Nature **508**(7495), 207–214 (2014). Tract-tracing study detailing the weights of connections received by 295 structures in the mouse brain. (A)

**38.** "Scale-free brain functional networks," V. M. Eguiluz, D. R. Chialvo, G. A. Cecchi, M. Baliki, and A. V. Apkarian, Phys. Rev. Lett. **94**(1), 018102 (2005). Analysis of fMRI recordings demonstrates that functional connectivity between brain regions exhibits scale-free, small-world structure. (A)

**39.** "Rich-club organization of the human connectome," M. P. Van Den Heuvel and O. Sporns, J. Neurosci. **31**(44), 15775–15786 (2011). Analyzes connectivity structure of the human brain obtained using diffusion tensor imaging data and identifies a sub-network of hub regions that are highly connected with one another. (A)

**40.** "Connectivity optimization and the positioning of cortical areas," V. A. Klyachko and C. F. Stevens, Proc. Natl. Acad. Sci. **100**(13), 7937–7941 (2003). Suggests that the spatial configuration of various functional areas in the cortex evolved to minimize the net axonal volume required to wire the cortex. (A)

## VI. NEURONAL SYNCHRONIZATION

Transient synchronization of neuronal spiking (i.e., action potential generation) is a ubiquitous feature of brain dynamics, being one strategy employed by upstream neuronal populations to reliably activate downstream populations and thereby transmit information. Elevated synchronization is associated with various higher-order cognitive functions, such as directed attention and memory encoding. Synchronization has also been proposed as a mechanism for transient communication between disparate brain areas,[43] a possibility that could explain how various sensory modalities are "bound" into unified perception.[44,65] In some situations, however, coordinated spiking may be harmful, as excessive synchrony has been implicated in diseases such as epilepsy, Parkinson's disease, and schizophrenia.[47]

Efforts to understand how neuronal networks synchronize have greatly benefited by borrowing tools from dynamical systems theory. By treating individual neurons as nonlinear oscillators, it is possible to define a neuron's Phase Response Curve (PRC), which describes the phase shift exhibited by a repetitively firing neuron when it receives a weak stimulus (such as a synaptic input from another neuron). It has been shown that biological neurons generally display one of two kinds of PRC's, with neurons with so-called Type II PRC's synchronizing better than neurons with Type I PRC's in excitatory networks.[46,48]

Aside from individual neuronal properties such as the PRC, the other major factor influencing neuronal synchrony is the pattern of connectivity by which neurons couple. As described above, methods from statistical physics have proven useful in characterizing neuronal network structure, and a combination of computational and theoretical analysis has begun to reveal how properties of networks influence collective synchronization. One particularly useful approach is Master Stability Function analysis,[49,50] which uses the spread in the eigenvalues of a network's connectivity matrix to characterize the network's propensity for synchronization.

**41.** "The language of the brain," T. Sejnowski and T. Delbruck, Sci. Am. **307**(4), 54–59 (2012). Gives concrete examples of how neuronal synchronization contributes to information processing in the brain, particularly in vision, attention, and memory. (E)

**42.** "Simulating synchronization in neuronal networks," C. G. Fink, Am. J. Phys. **84**(6), 467–473 (2016). Reviews fundamental models and numerical methods for simulating neuronal networks, as well as quantifying their level of synchronization. (E)

**43.** "The brainweb: Phase synchronization and large-scale integration," F. Varela, J. P. Lachaux, E. Rodriguez, and J. Martinerie, Nat. Rev. Neurosci. **2**(4), 229–239 (2001). Reviews experimental evidence for synchronization being essential to integrating information from various parts of the brain. (I)

**44.** "The temporal correlation hypothesis of visual feature integration: Still alive and well," C. M. Gray, Neuron **24**(1), 31–47 (1999). Summarizes the "binding problem" with respect to vision and describes how neuronal synchronization may provide a solution. (I)

**45.** "Neurons as oscillators," K. M. Stiefel and G. B. Ermentrout, J. Neurophysiol. **116**(6), 2950–2960 (2016). Review of neuronal phase-response curve analysis, with an emphasis on biological applicability. (I)

**46.** "Analysis of neural excitability and oscillations," J. Rinzel and B. Ermentrout, in **Methods in Neuronal Modeling**, edited by C. Koch, and I. Segev (MIT Press, Cambridge, MA, 1998), pp. 251–292. Analysis of different dynamics resulting from Morris-Lecar model neurons with Type I versus Type II phase response curves. (I)

**47.** "Neural synchrony in brain disorders: relevance for cognitive dysfunctions and pathophysiology," P. H. Uhlhaas and W. Singer. Neuron **52**(1), 155–168 (2006). Review of evidence associating neuronal synchronization with brain pathologies such as epilepsy, Parkinson's disease, Alzheimer's disease, autism, and schizophrenia. (I)

**48.** "Synchrony in excitatory neural networks," D. Hansel, M. Germán, and C. Meunier, Neural Comput. **7**(2), 307–337 (1995). One of the earliest papers applying PRC analysis to neuronal network dynamics. (A)

**49.** "Master stability functions for synchronized coupled systems," L. M. Pecora and T. L. Carroll, Phys. Rev. Lett. **80**(10), 2109–2112 (1998). Introduces the concept

of master stability function analysis to mathematically separate the effects of network topology from individual oscillator dynamics in determining network synchronization. (A)

**50.** "Enhancement of neural synchrony by time delay," M. Dhamala, V. K. Jirsa, and M. Ding, Phys. Rev. Lett. **92**(7), 074104 (2004). Uses the master stability function approach to show that delays in coupling between neurons may enhance network synchronizability. (A)

**51.** "Transition from local to global phase synchrony in small world neural network and its possible implications for epilepsy," B. Percha, R. Dzakpasu, M. Żochowski, and J. Parent, Phys. Rev. E **72**(3), 031909 (2005). Demonstrates that global phase synchronization of a simulated neuronal network with Watts-Strogatz connectivity dramatically increases as the network is re-wired with long-range "shortcut" connections. (A)

**52.** "GABAergic hub neurons orchestrate synchrony in developing hippocampal networks," P. Bonifazi *et al.*, Science **326**(5958), 1419–1424 (2009). Experimental results indicating that highly connected "hub" neurons control neuronal network synchronization in some cases. (A)

**53.** "Network synchronization in hippocampal neurons," Y. Penn, M. Segal, and E. Moses, Proc. Natl. Acad. Sci. **113**(12), 3341–3346 (2016). Experimental results showing that cultured neurons act as intrinsic oscillators whose synchronization increases as coupling strength increases, matching theoretical predictions. (A)

## VII. STOCHASTIC AND COHERENCE RESONANCE

While theoretically treating neurons as intrinsic oscillators has been very fruitful (as described in Sec. VI), this assumption is not always justified. Even though neurons are nonlinear oscillators that may spike regularly when sufficiently stimulated, neuronal activity recorded from freely behaving animals indicates that sparse, irregular firing is more commonly observed than periodic, network-wide "bursting." The ubiquity of such irregular dynamics within the inherently noisy environment common to any biological system suggests that the brain has evolved strategies not merely for tolerating noise, but for harnessing it to improve information transmission.

One mechanism for doing so is through stochastic resonance, a concept initially proposed to explain Earth's cycles of ice ages and which has since found application in many other scientific fields. In a system that displays stochastic resonance, a weak signal of interest is more easily detected by the addition of noise, with an optimal noise amplitude producing the best detectability. The simplest example in neuroscience would be a single sensory neuron whose membrane potential undergoes a subthreshold oscillation due to a particular stimulus, with spikes occurring due to voltage fluctuations caused by noise.

Coherence resonance is a related concept, with the difference being that coherence resonance includes no concept of an input signal. In our single-neuron analogy, rather than responding to a stimulus with sub-threshold oscillations, the neuron is simply at rest, and a certain level of perturbing noise results in the most nearly periodic spiking of the neuron. (This is in fact how the concept of coherence resonance was initially proposed—see Ref. 60 below.)

**54.** "What is stochastic resonance? Definitions, misconceptions, debates, and its relevance to biology," M. D. McDonnell and D. Abbott, PLoS Comput. Biol. **5**(5), e1000348 (2009). Introduction to the concept of stochastic resonance, with a focus on applications in neuroscience. (E)

**55.** "The benefits of noise in neural systems: Bridging theory and experiment," M. D. McDonnell and L. M. Ward, Nat. Rev. Neurosci. **12**(7), 415–426 (2011). A more thorough review of the theoretical and experimental studies investigating stochastic resonance in neuronal systems. (I)

**56.** "Time-interval sequences in bistable systems and the noise-induced transmission of information by sensory neurons," A. Longtin, A. Bulsara, and F. Moss, Phys. Rev. Lett. **67**, 656–659 (1991). One of the first studies to suggest that stochastic resonance may be harnessed by individual sensory neurons to transmit information. (I)

**57.** "Noise enhancement of information transfer in crayfish mechanoreceptors by stochastic resonance," J. K. Douglass, L. Wilkens, E. Pantazelou, and F. Moss, Nature **365**(6444), 337–339 (1993). One of the first studies to experimentally demonstrate stochastic resonance in a neural system. (I)

**58.** "Stochastic resonance in a neuronal network from mammalian brain," B. J. Gluckman, T. I. Netoff, E. J. Neel, W. L. Ditto, M. L. Spano, and S. J. Schiff, Phys. Rev. Lett. **77**(19), 4098–4101 (1996). Experimental demonstration of stochastic resonance observed by recording the extracellular potential from hippocampal slices. (I)

**59.** "Delay-induced multiple stochastic resonances on scale-free neuronal networks," Q. Wang, M. Perc, Z. Duan, and G. Chen, Chaos **19**(2), 023112 (2009). Computational study demonstrating that a neuronal network with scale-free connectivity exhibits stochastic resonance when a pacemaker cell drives one other cell in the network. (A)

**60.** "Coherence resonance in a noise-driven excitable system," A. S. Pikovsky and J. Kurths, Phys. Rev. Lett. **78**(5), 775–778 (1997). This study introduced the concept of coherence resonance, using the Fitzhugh-Nagumo model neuron as its paradigmatic example. (A)

**61.** "Noise-induced cooperative dynamics and its control in coupled neuron models," B. Hauschildt, N. B. Janson, A. Balanov, and E. Schöll, Phys. Rev. E **74**(5), 051906 (2006). Explores the effect of delayed feedback on coherence resonance observed in a system of two coupled Fitzhugh-Nagumo neurons. (A)

**62.** "Collective stochastic coherence in recurrent neuronal networks," B. Sancristóbal, B. Rebollo, P. Boada, M. V. Sanchez-Vives, and J. Garcia-Ojalvo, Nat. Phys. **12**(9), 881–887 (2016). Experimental demonstration of coherence resonance in slices from visual cortex. (A)

## XIII. EXTRACELLULAR RHYTHMS

Ever since Hans Berger recorded the first electroencephalogram (EEG) in the 1920s,[63] scientists have been fascinated by the electrical undulations perpetually produced by the brain. Clinicians have categorized such recordings based on their peak spectral content: delta waves (traditionally 0.5–4 Hz) are prominent in the cortex during deep sleep, theta waves (4–8 Hz) are observed in the hippocampus

during environmental exploration, alpha waves (8–12 Hz) are seen in the occipital cortex when a subject's eyes are closed, beta waves (12–30 Hz) are prominent in patients with Parkinson's disease, and gamma waves (>30 Hz) are associated with selective visual attention (to give just a few examples). The scientist naturally wonders if there is any principled reason for categorizing neuronal oscillations in this way. What biophysical processes underpin these various rhythms? Moreover, what is the origin of these electrical signals in the first place?

These questions physicists have the tools to answer. To good approximation, the existence of detectable extracellular electric potentials results from volume conduction of currents across neuronal membranes. Combining the quasi-static approximation of Maxwell's equations with Ohm's law, one may calculate the electric potential at a recording electrode (which when positioned inside the brain records the *local field potential*, or LFP) due to known transmembrane currents. (Going backward from LFP to currents is an ill-posed inverse problem, though many heuristic techniques have been developed.)

An abundance of biophysical processes generates the zoo of observed brain rhythms. Gamma rhythms have received the most attention, since they are theorized to aid in communication between disparate brain regions (see Ref. 43), perhaps even playing a pivotal role in generating consciousness. It has been shown that neuronal networks may produce gamma rhythms through strictly inhibitory coupling, through excitatory-inhibitory feedback loops (so-called "PING": pyramidal-inhibitory-gamma), and through more exotic mechanisms.[68]

A more fundamental mechanistic question is how sparse and/or asynchronous neural activity may generate narrowband LFP oscillations. While it is intuitive that a network of synchronous, periodically firing neurons will do so, many experiments have shown clear rhythms associated with seemingly stochastic, aperiodic firing of individual cells. This remains an ongoing area of investigation.[73,74]


**63.** "Über das elektrenkephalogramm des menschen," H. Berger, Arch. Psych. Nervenkrankheiten **87**(1), 527–570 (1929).

**64.** **Rhythms of the Brain**, G. Buzsaki (Oxford U.P., Oxford, 2006). A thorough introduction to the diversity of brain rhythms, the functions with which they are associated, and what is known of their biophysical underpinnings. Written at a popular level with technical details included in the copious footnotes. (E)

**65.** "Binding by synchrony," W. Singer, http://www.scholarpedia.org/article/Binding_by_synchrony **2**(12), 1657 (2007). Overview of the idea that neuronal oscillations are essential to "binding" information from various parts of the brain, as happens when the brain's representations of visual features (for example) are combined to generate a unified perception of a complex object (such as a face). (E)

**66.** "The origin of extracellular fields and currents—EEG, ECoG, LFP and spikes," G. Buzsáki, C. A. Anastassiou, and C. Koch, Nat. Rev. Neurosci. **13**(6), 407–420 (2012). Reviews the various methods for recording extracellular potentials, the biophysics of how they are generated, and methods for estimating neuronal currents from extracellular recordings (I).

**67.** "LFPy: A tool for biophysical simulation of extracellular potentials generated by detailed model neurons," H. Lindén *et al.*, Front. Neuroinf. **7**, 1–15 (2013). Describes a computational tool for simulating extracellular potentials generated by neuronal activity, using two different mathematical methods: the point-source and line-source approximations, both of which assume the quasi-static approximation of Maxwell's equations. (I)

**68.** "Mechanisms of gamma oscillations," G. Buzsáki and X. J. Wang, Annu. Rev. Neurosci. **35**, 203–225 (2012). Review of the various biophysical mechanisms that generate gamma oscillations in the brain. (I)

**69.** "Perception's shadow: Long-distance synchronization of human brain activity," E. Rodriguez *et al.*, Nature **397**(6718), 430–433 (1999). Presents evidence for large-scale gamma oscillations observed in subjects' EEG recordings the moment they perceive a face in an ambiguous image. (I)

**70.** "From Maxwell's equations to the theory of current-source density analysis," S. L. Gratiy *et al.*, Eur. J. Neurosci. **45**(8), 1013–1023 (2017). Discusses the applicability of the quasi-static approximation of Maxwell's equations to extracellular potentials and determines the conditions under which extracellular diffusion and other currents must be considered (in addition to transmembrane currents) in order to obtain accurate estimates of transmembrane current densities from LFP recordings. (A)

**71.** "Kernel current source density method," J. Potworowski, W. Jakuczun, S. Łęski, and D. Wójcik, Neural Comput. **24**(2), 541–575 (2012). Presents an "inverse" method for deducing transmembrane currents from the recorded LFP signal. (A)

**72.** "Gamma oscillation by synaptic inhibition in a hippocampal interneuronal network model," X. J. Wang and G. Buzsáki, J. Neurosci. **16**(20), 6402–6413 (1996). Computationally demonstrates that gamma rhythms may be generated by a population of neurons that are coupled purely through inhibition. (A)

**73.** "What determines the frequency of fast network oscillations with irregular neural discharges? I. Synaptic dynamics and excitation-inhibition balance," N. Brunel and X. J. Wang, J. Neurophysiol. **90**(1), 415–430 (2003). Theoretical explanation of how high-frequency rhythms may be generated by neurons firing sporadically at much lower frequencies. (A)

**74.** "Emergence of narrowband high frequency oscillations from asynchronous, uncoupled neural firing," S. V. Gliske, W. C. Stacey, E. Lim, K. A. Holman, and C. G. Fink, Int. J. Neural Syst. **27**(1), 1650049 (2017). Derives how narrowband oscillations may be generated by populations of neurons firing asynchronously, but with similar firing frequencies. (A)

**75.** "Prediction of electroencephalographic spectra from neurophysiology," P. A. Robinson *et al.*, Phys. Rev. E **63**(2), 021903 (2001). Proposes a theoretical model to explain the observed characteristics of EEG power spectra and attempts to provide a basis for breaking down EEG spectra into physiologically meaningful frequency bands. (A)


## IX. CRITICALITY AND CHAOS IN NEURONAL DYNAMICS

One tantalizing theory proposed in the last few decades is that the brain is a critical system, with finely tuned system

parameters enabling unexpected macroscopic properties to emerge from microscopic elements. Just as a single spin-flip in the Ising model near the critical temperature may provoke an avalanche that flips the entire system's magnetic moment, so too might a single neuronal action potential reshape the dynamics of an entire brain (and corresponding behavior of the organism). This idea provides an elegant explanation for the brain's ability to adapt with such agility to manifold external stimuli.

One hallmark of critical dynamics is scale invariance (i.e., a power-law relationship between a measure of interest and some system parameter), either in space or in time. "Neuronal avalanches" have been experimentally observed to exhibit scale-free distributions both in avalanche size and duration.[81] In addition, analysis of whole-brain functional magnetic resonance imaging (fMRI) recordings reveals a scale invariant correlation structure virtually identical to that of the 2D Ising model at the critical temperature.[82] While such data do not prove that the brain is a critical system, they are consistent with the hypothesis, and they continue to spur research into the possibility that brain dynamics may follow principles universal to many other physical systems.

A related line of inquiry has investigated whether the brain exhibits chaotic dynamics (featuring sensitive dependence upon initial conditions, a characteristic distinct from the long-range correlations associated with critical dynamics). A chaotic attractor with multiple "wings" (picture the Lorenz butterfly) could in principle underpin information processing if different inputs pushed brain dynamics into different "wings"[86] (similar to a multistable system, as described in Sec. X). While chaos has been experimentally demonstrated at the single-neuron level, it has proven much more difficult to identify chaotic dynamics in macroscopic brain signals such as EEG. Many studies in the 1980s and early 1990s claimed to find such evidence, but stringent analysis overturned these results. To date there is little evidence of chaos in EEG associated with healthy cortical dynamics. There are, however, plausible indications of chaotic dynamics in EEG recorded during epileptic seizures.[85]


**76.** "Resource Letter CPPPT-1: Critical point phenomena and phase transitions," J. Tobochnik, Am. J. Phys. **69**, 255–263 (2001). Overview of the literature related to critical phenomena. (E)

**77.** "Emergent complex neural dynamics," D. R. Chialvo, Nat. Phys. **6**(10), 744–750 (2010). Review of fundamental concepts related to criticality, as well as of signature studies suggesting that the brain functions near criticality. (E)

**78.** "The functional benefits of criticality in the cortex," W. L. Shaw and D. Plenz, The Neuroscientist **19**(1), 88–100 (2013). Review presents a toy model of criticality in a neural system, then describes work suggesting that information transmission and information capacity are optimized in neuronal networks operating at criticality. (E)

**79.** "Self-organized criticality as a fundamental property of neural systems," J. Hesse and T. Gross, Front. Syst. Neurosci. **8**(166), 1–14 (2014). Review of the criticality hypothesis that includes an extensive discussion of models of *self-organized* criticality in the brain. (I)

**80.** "Critical branching captures activity in living neural networks and maximizes the number of metastable states," C. Haldeman and J. M. Beggs, Phys. Rev. Lett. **94**(5), 058101 (2005). Proposes a model of how neuronal activation spreads throughout an experimental neuronal network and shows that the number of metastable states (useful for information processing) is maximized at criticality. (I)

**81.** "Neuronal avalanches in neocortical circuits," J. M. Beggs and D. Plenz, J. Neurosci. **23**(35), 11167–11177 (2003). Seminal work demonstrating scale-free distributions in size and duration of "neuronal avalanches" recorded from cultured neocortical neurons. (A)

**82.** "Ising-like dynamics in large-scale functional brain networks," D. Fraiman, P. Balenzuela, J. Foss, and D. R. Chialvo, Phys. Rev. E **79**(6), 061922 (2009). This study shows that fMRI data recorded from resting human brains exhibit nearly identical correlation structure to the 2D Ising model at the critical temperature. (A)

**83.** "Brain organization into resting state networks emerges at criticality on a model of the human connectome," A. Haimovici, E. Tagliazucchi, P. Balenzuela, and D. R. Chialvo, Phys. Rev. Lett. **110**(17), 178101 (2013). Simulating critical brain dynamics on a network with anatomically realistic connectivity reproduces several key experimental findings. (A)

**84.** "Does the 1/f frequency scaling of brain signals reflect self-organized critical states?," C. Bedard, H. Kroeger, and A. Destexhe, Phys. Rev. Lett. **97**(11), 118102 (2006). Proposes a model for how brain dynamics may exhibit $1/f$ frequency scaling without appealing to critical dynamics. (A)

**85.** "Dynamic models of large-scale brain activity," M. Breakspear, Nat. Neurosci. **20**(3), 340–352 (2017). Gives a historical overview of efforts to find evidence of chaos in EEG signals, and reviews various macroscopic models of brain dynamics. (E)

**86.** "Is there chaos in the brain? II. Experimental evidence and related models," H. Korn and P. Faure, C.R. Biol. **326**(2), 787–840 (2003). A comprehensive overview of the search for chaos in brain dynamics at various spatial scales. (I)

**87.** "Comparison of EEG time series from rat olfactory system with model composed of nonlinear coupled oscillators," L. Kay, K. Shimoide, and W. J. Freeman, Int. J. Bifur. Chaos **5**(3), 849–858 (1995). Presents a computational model of chaotic dynamics in the rat olfactory cortex and compares model results with experimental data. The last author, Walter Freeman, was one of the primary pioneers in applying chaos theory to the brain. (A)


## X. COORDINATION DYNAMICS

The theory of Coordination Dynamics proposes a mechanism by which the brain may harness criticality to process information: By resting at a critical point, different sensory perturbations may push the brain's dynamics into different attractor states. As a toy example, the sight of food may push a brain's dynamics into an attractor state that results in foraging behavior, while the sight of a snake may switch the dynamics to induce an escape response. It is assumed the various attractors are stable, so that a perturbation of some sort is required to switch states. This is why this paradigm is known as the *multistable* theory of brain dynamics.[90] Different attractor states are posited to span multiple brain regions, with phase-locking synchrony between these regions somehow integrating their information. While this "binding

by synchrony" idea has been proposed in other theories of brain function,[43,65] the theory of Coordination Dynamics emphasizes that segregation of brain modules through *de*synchronization should be just as important as their integration through synchronization. (This emphasis on integration and segregation is similar to the perspective articulated by Tononi *et al.* in Ref. 33, except from a dynamical systems perspective, rather than information theoretic.)

The Coordination Dynamics paradigm suggests that a different dynamical regime may help to explain the tension between synchronous and asynchronous activity observed in the brain. In place of the landscape of stable attractors inherent in *multi*stable dynamics, the brain may, at times, function in a *meta*stable regime completely devoid of stable states. Metastable dynamics are characterized by "attractive states" that generate brief periods of quasisynchrony, but such states are intrinsically unstable and naturally give way to segregated (asynchronous) dynamics even in the absence of external perturbation. Metastability naturally emerges in weakly coupled systems composed of heterogeneous oscillators, and several studies have produced electrophysiological data consistent with such dynamics.


**88.** "Haken-Kelso-Bunz model," J. A. S. Kelso, http://www.scholarpedia.org/article/Haken-Kelso-Bunz_model, Scholarpedia, **3**(10), 2008. Simple mathematical model of a prototypical metastable system. This model was originally used to describe the transition from anti-phase to in-phase finger tapping observed in behavioral experiments. (E)

**89.** "The metastable brain," E. Tognoli and J. A. S. Kelso, Neuron **81**(1), 35–48 (2014). Review of behavioral and electrophysiological studies supporting the idea that the brain may function as a metastable system. (I)

**90.** "Multifunctional pattern-generating circuits," K. L. Briggman and W. B. Kristan, Jr., Annu. Rev. Neurosci. **31**, 271–294 (2008). Reviews experimental evidence that the brain may function as a multistable system. (I)

**91.** "Brain coordination dynamics: True and false faces of phase synchrony and metastability," E. Tognoli and J. A. S. Kelso, Prog. Neurobiol. **87**(1), 31–40 (2009). Presents evidence for transient synchronization and desynchronization between multiple brain regions in human EEG data and offers a model for disentangling the confounding effects of volume conduction to identify true synchronization between disparate brain regions. (A)

**92.** "Resting-state temporal synchronization networks emerge from connectivity topology and heterogeneity," A. Ponce-Alvarez e*t al*., PLoS Comput. Biol. **11**(2), e1004100 (2015). Presents evidence for transient synchronization and desynchronization of multiple brain regions in human fMRI data, and formulates a computational model suggesting that their experimental data are consistent with metastable dynamics. (A)


## XI. POPULATION MODELS AND WHOLE-BRAIN SIMULATIONS

When modeling interactions between multiple neuronal populations, it is often computationally infeasible to simulate individual neurons. Instead, a large number of neurons may be collectively modeled by introducing a population density, for example, $p(v,t)$, which when integrated gives the fraction of neurons whose membrane potentials are between $v$ and $v+dv$ within a temporal window. Standard techniques from statistical physics approximate the evolution of $p(v,t)$ in terms of flow (i.e., the intrinsic dynamics of the individual neurons) and diffusion (due to the inherent stochasticity in any biological system), which together can be described using a Fokker-Planck formalism. Mean-field analysis then uses equilibrium solutions of the Fokker-Planck equation to derive the transfer function of a neural population, describing the mean firing rate that results from a given level of either synaptic current input or membrane depolarization.[94]

Neural mass models then employ transfer functions to define the coupling interactions between multiple neuronal populations, assuming that the mean firing rate of one neuronal population is affected only by the mean (as opposed to variance, or higher-order moments) firing rates of other populations.[85,94] This approach has been particularly useful for investigating how interactions between brain regions generate the rhythms and evoked responses observed in EEG recordings.[95] Neural field models go a step beyond this, modeling spatiotemporal pattern generation by considering population-averaged voltage as a function of continuous space and time. Neural field models have been used to explore spatiotemporal pattern formation[96] and traveling waves of neuronal activity[97] in brain tissue.

The computational efficiency gained by these coarse-grained models has enabled simulation of whole-brain dynamics. Such highly simplified brain models are inherently limited in the questions they can address, but they have proven useful in exploring how large-scale structural connections give rise to observed correlations between various brain regions (i.e., "functional connectivity"). There is also hope that such simulations can be tailored to individual patient's brains to treat "dynamical diseases" such as epilepsy.[98]


**93.** "Wilson-Cowan model," Z. P. Kilpatrick, Encyclopedia of Computational Neuroscience, 3159–3163 (2015). Overview of the original neural mass/neural field model, which features a sigmoidal function relating firing rate to membrane potential and includes limit cycle solutions, which model brain rhythms. (E)

**94.** "The dynamic brain: From spiking neurons to neural masses and cortical fields," G. Deco, V. K. Jirsa, P. A. Robinson, M. Breakspear, and K. Friston, PLoS Comput. Biol. **4**(8), e1000092 (2008). Rigorously yet succinctly describes mean field models, neural mass models, and neural field models, and reviews their contributions to our understanding of brain function. (I)

**95.** "Modelling event-related responses in the brain," O. David, L. Harrison, and K. J. Friston, NeuroImage **25**(3), 756–770 (2005). Uses a neural mass model to explore how evoked responses in EEG and MEG recordings depend on properties of the connectivity between brain regions. (A)

**96.** "Dynamics of pattern formation in lateral-inhibition type neural fields," S. Amari, Biol. Cybernet. **27**(2), 77–87 (1977). Presents a neural field model in which connectivity is excitatory between nearby brain regions and inhibitory between more distant brain regions. Demonstrates that such connectivity allows for stimulation of a localized area to induce an "activity bubble" that remains active even after stimulation ceases. Such persistent activity has been linked to working memory. (A)

**97.** "Control of traveling waves in the mammalian cortex," K. A. Richardson, S. J. Schiff, and B. J. Gluckman, Phys. Rev. Lett. **94**(2), 028103 (2005). Experimentally verifies a key prediction made by neural field models concerning traveling waves in brain tissue. (A)

**98.** "The virtual brain: A simulator of primate brain network dynamics," P. S. Leon *et al.*, Front. Neuroinf. **7**, (2013). Describes the development and capabilities of a freely available computing platform to simulate whole-brain dynamics. (I)

**99.** "Emerging concepts for the dynamical organization of resting-state activity in the brain," G. Deco, V. K. Jirsa, and A. R. McIntosh, Nat. Rev. Neurosci. **12**(1), 43–56 (2011). Reviews various models of whole-brain dynamics that attempt to explain observed patterns of functional connectivity, as well as to understand how these patterns of correlated brain activity underpin brain function. (I)

**100.** "Key role of coupling, delay, and noise in resting brain fluctuations," G. Deco, V. Jirsa, A. R. McIntosh, O. Sporns, R. Kötter, and M. E. Raichle, Proc. Natl. Acad. Sci. U.S.A. **106**(25),10302–10307 (2009). Employs the Wilson-Cowan model to simulate the macaque monkey brain, exploring the physiological parameters for which their model results best match fMRI-derived resting-state functional connectivity data. (A)

## XII. METHODS OF RECORDING NEURONAL ACTIVITY AND STRUCTURE

Many contemporary methods for noninvasively ascertaining neuronal connectivity and recording brain activity are applications of fundamental physical principles. Functional magnetic resonance imaging (fMRI), for example, is predicated on the fact that the deoxygenation of blood alters its magnetic susceptibility, which allows nuclear magnetic resonance (NMR) imaging to identify areas of the brain which receive increased blood flow (which serves as a useful proxy for increased neuronal activity).

NMR can also track the Brownian motion of water molecules, which are restricted to diffusing primarily along axonal fibers within the white matter of the cerebral cortex. By determining the diffusion tensor as a function of spatial location throughout the cortex, it is possible to reconstruct axonal fiber tracts and thereby deduce cortical connectivity structure. Diffusion tensor imaging (along with its upgraded cousins, diffusion spectrum imaging and high-angular resolution diffusion imaging) is currently our best approach for noninvasively investigating human brain connectivity.

One up-and-coming imaging technology is similar to fMRI in exploiting differences between oxygenated and deoxygenated hemoglobin. While fMRI takes advantage of these molecules' distinct magnetic properties, functional near infrared imaging (fNIRI) uses molecules' different absorption and scattering spectra. By using an array of emitters and detectors to transmit near infrared light through the skull and to measure the intensity of reflected light that emerges, a crude map of blood flow in the brain can be reconstructed (assumed to correlate with neuronal activity). While the resolution and reliability of this technique fall far short of fMRI, fNIRI systems are comparatively low-cost and portable.

Another excellent noninvasive tool is positron emission tomography, in which a radioactive tracer which undergoes $\beta^+$ decay (such as carbon-11 or fluorine-18) is incorporated into a molecule of interest, and a sample containing the compound is administered internally. Gamma rays, which result from the emitted positrons encountering electrons, are then detected and used to image the location of the molecules. For instance, in studies of Alzheimer's disease it is often necessary to detect and locate pathological beta-amyloid protein fragments. PET scans can accomplish this by using the molecule florbetaben, which incorporates fluorine-18 and binds to beta-amyloid plaques.

One final non-invasive tool worth mentioning is magnetoencephalography (MEG), which measures the magnetic field generated by neuronal currents (rather than the electric field, as in EEG). The primary advantage of MEG over EEG is that the skull is magnetically transparent, so that MEG signals do not suffer the same attenuation and distortion as EEG signals. Still, neuronal currents are so small, between 10 and 100 femtotesla, that superconducting quantum interference devices (SQUID's) are required to measure the MEG signals. A typical MEG device consists of over 300 SQUID's which must be cooled to 4 K by liquid helium. MEG is therefore a much more expensive and cumbersome tool than EEG. MEG is still useful in many cases, however, because it is sensitive to different orientations of neuronal current than EEG, so that the two signals offer complementary information about large-scale brain activity.

While the foregoing non-invasive techniques have all played an important role in furthering our understanding of the brain, they lack the simultaneous spatial and temporal resolution necessary to track the activity of millions of individual neurons (as do invasive approaches, such as two-photon microscopy and tetrode recordings). The United States' BRAIN Initiative (Brain Research through Advancing Innovative Neurotechnologies) has earmarked funding to develop such technologies, which are necessary to formulate a framework for how specific brain circuits operate. Spurred by this initiative, many physicists are working to develop tools to generate data that will inspire the next revolution in our understanding of the brain.

**101.** "Physics of MRI: A primer," D. B. Plewes and W. Kucharczyk, J. Mag. Reson. Imaging **35**(5), 1038–1054 (2012). Conceptual introduction to the physics of nuclear magnetic resonance imaging. (E)

**102.** "White's the matter," K. Rae Chi, Scientist **28**(11), 67–69 (2014). Explains the differences between diffusion tensor imaging, diffusion spectrum imaging, and high-angular resolution diffusion imaging. (E)

**103.** "The physics of functional magnetic resonance imaging (fMRI)," R. B. Buxton, Rep. Prog. Phys. **76**(9), 096601 (2013). Thorough review of the physical basis for our ability to use NMR to detect changes in the blood oxygen level dependent (BOLD) signal throughout the brain. Includes a discussion of the care that must be taken in relating the BOLD signal to neuronal activity. (I)

**104.** "Diffusion tensor MR imaging and fiber tractography: Theoretic underpinnings," P. Mukherjee, J. I. Berman, S. W. Chung, C. P. Hess, and R. G. Henry, Am. J. Neuroradiol. **29**(4), 632–641 (2008). Conceptual overview of diffusion tensor imaging. (I)

**105.** "Physical foundations, models, and methods of diffusion magnetic resonance imaging of the brain: A

review," L. Minati and W. P. Węglarz, Concepts Magn. Reson. Part A **30**(5), 278–307 (2007). Rigorous overview of the physics and the mathematical methods underpinning diffusion tensor imaging. (A)

**106.** "Nuclear techniques for medical imaging: Positron emission tomography," M. A. Mandelkern, Annu. Rev. Nucl. Part. Sci. **45**(1), 205–254 (1995). Overview of the core physics and mathematical techniques involved in PET imaging. (I)

**107.** "Advantages in functional imaging of the brain," W. Mier and D. Mier, Front. Human Neurosci. **9**, 1–6 (2015). Gives examples of applications of PET imaging in basic brain research and discusses how PET and fMRI complement one another. (I)

**108.** "A review on continuous wave functional near-infrared spectroscopy and imaging instrumentation and methodology," F. Scholkmann *et al.*, Neuroimage **85**, 6–27 (2014). Thorough review of the physics, algorithms, and hardware used in fNIRI. (I)

**109.** "Magnetoencephalography: From SQUIDs to neuroscience: Neuroimage 20th anniversary special edition," R. Hari and R. Salmelin, Neuroimage **61**(2), 386–396 (2012). General overview of the biophysical basis for detecting MEG signals, as well as the most important historical applications of this technique. (I)

**110.** "The brain is big science," T. Feder, Phys. Today **66**(12), 20–22 (2013). An overview of the U.S. BRAIN Initiative and the technologies under development, as well as a summary of the early days of the EU's Human Brain Project, which initially sought to simulate the entire human brain. (E)

**111.** "BRAIN 2025: A Scientific Vision," C. Bargmann *et al.*, Brain Research Through Advancing Innovative Neurotechnologies (BRAIN) Working Group Report to the Advisory Committee to the Director, NIH. Available online at: https://www.braininitiative.nih.gov/pdf/BRAIN2025_508C.pdf (U.S. National Institutes of Health, 2014). Outline the BRAIN Initiative's goals. (I)

**112.** "Direct wavefront sensing for high-resolution in vivo imaging in scattering tissue," K. Wang, W. Sun, C. T. Richie, B. K. Harvey, E. Betzig, and N. Ji, Nat. Commun. **6**, 1–6 (2015). Applies adaptive optics techniques borrowed from astrophysics to improve two-photon imaging of the mouse brain. (A)

**113.** "In vivo three-photon imaging of activity of GCaMP6-labeled neurons deep in intact mouse brain," D. G. Ouzounov *et al.*, Nat. Methods **14**, 388–390 (2017). Describes an improvement over two-photon microscopy for imaging neuronal activity in a living organism. (A)

## XIII. NEUROMORPHIC COMPUTING

Simulating the dynamics of neuronal networks on digital computers is terribly slow and incredibly inefficient, requiring orders of magnitude more time and consuming orders of magnitude more energy than the same number of neurons in the brain. This is because the fundamental components of a neuron—a "capacitor" (cell membrane) and many "resistors" (ion channels)—are analog, so that simulating their dynamics with on/off transistors is cumbersome.

The desire to improve both speed and efficiency of neuronal simulations has spawned the field of "neuromorphic computing," which aims to design new generations of computing hardware whose fundamental components take their cues from the brain. For example, transistors may function below their turn-on thresholds to mimic analog synapses, and neuromorphic chips may compute "asynchronously," running many processes in parallel without a master clock. Efforts to develop neuromorphic chips are being spearheaded not only by corporations such as IBM and Intel, but also by the European Union-funded "Human Brain Project," which is developing a variety of tools within computational neuroscience.

One particularly promising development in the last decade has been the fabrication of memristors, which are fundamental circuit components whose resistance depends upon the applied voltage and current over time. This is similar to neuronal synapses, whose conductances may vary depending upon their use (recall that such "neuronal plasticity" is essential to memory storage). Much effort is currently concentrated on incorporating memristors into neuromorphic chips in order to simulate neuronal learning at the hardware level.

**114.** "Why the Human Brain Project went wrong—and how to fix it," S. Theil, Sci. Am. **313**(4), 36–42 (2015). Describes the difficulties experienced in the first few years of the Human Brain Project. (E)

**115.** "Worldwide initiatives to advance brain research," S. Grillner *et al.*, Nat. Neurosci. **19**(9), 1118–1122 (2016). Includes a short synopsis of the current state of the Human Brain Project (as well as five other large-scale neuroscientific endeavors, including the BRAIN Initiative). (E)

**116.** "Large-scale neuromorphic computing systems," S. Furber, J. Neural Eng. **13**(5), 051001 (2016). Reviews the history of neuromorphic computing and evaluates the approaches of several large-scale projects currently underway. (I)

**117.** "Memristive devices for computing," J. J. Yang, D. B. Strukov, and D. R. Stewart, Nat. Nanotechnol. **8**(1), 13–24 (2013). Comprehensive review of how memristive devices may be used in computing, including examples of applications in neuromorphic hardware. (I)

**118.** "Neuromorphic hardware in the loop: Training a deep spiking network on the brainscales wafer-scale system," S. Schmitt *et al.*, arXiv preprint: 1703.01909 (2017). Describes learning in a neuromorphic chip (developed by the physicist Karlheinz Meier, in conjunction with the Human Brain Project) that runs 10,000 times faster than an equivalent biological neuronal network. (A)

## XIV. QUANTUM MECHANICS AND THE BRAIN

One area of research that has the potential to upend our understanding of the brain is the intersection of quantum mechanics and neuroscience. While there is no shortage of unfounded speculation about how quantum mechanics could play a role in brain function (such as Roger Penrose's proposal that neuronal microtubules might attain a collective quantum coherent state[119] ), in the past twenty years several experimental and phenomenological studies have lent plausibility to the role of quantum mechanics in one particular neuronal system: the avian compass.

The European robin can sense the inclination of Earth's magnetic field, though exactly how it does so is unclear. While there is evidence that magnetite crystals are housed

within its beak, recent experiments cast doubt on their involvement in the bird's magnetic compass, which has been shown to depend on exposure to blue-green light. An alternative theory posits that a photon impinging on cryptochrome, a photoreceptor molecule in the robin's retina, results in electron transfer forming a "radical pair." The unpaired electrons (one in the valence shell of each radical molecule) are initially entangled in a singlet state, but an external magnetic field may induce transitions to triplet states (with the transition probability depending on the field's orientation). Crucially, the singlet and triplet states are hypothesized to generate different chemical products, which ultimately (through some unknown mechanism) result in different signals being sent to the bird's brain.

This theoretical "radical pair mechanism" received a boost in 2004, when experiments demonstrated that a magnetic field oscillating at 7 MHz disrupted robins' magnetoreception.[124] This frequency was far higher than the 100 kHz believed to be the maximum frequency that could possibly disrupt a magnetite-based avian compass. However, another study in 2014 observed that ambient urban electromagnetic noise was sufficient to disrupt the robin's magnetic compass, a surprising result that does not strongly favor the radical pair theory over the magnetite theory.[125] It therefore remains unclear whether quantum mechanics plays an essential role in the avian compass, though the idea is promising enough that it remains an active, exciting field of investigation.


**119. Shadows of the Mind**, R. Penrose (Oxford U.P., Oxford, 1994). Penrose draws on the ideas of anesthesiologist Stuart Hameroff to suggest that microtubules (polymers that form the cytoskeletal scaffolding of many cells, including neurons) might attain a globally coherent quantum mechanical state that endows the brain with computational abilities surpassing those of any Turing machine. (E)

**120.** "The butterfly, the fruit fly and the quantum robin," J. McFadden and J. Al-Khalilin **Life on the Edge: The Coming of Age of Quantum Biology** (Crown Publishers, 2016), pp. 166–195. Popular account of the development of the radical pair theory. (E)

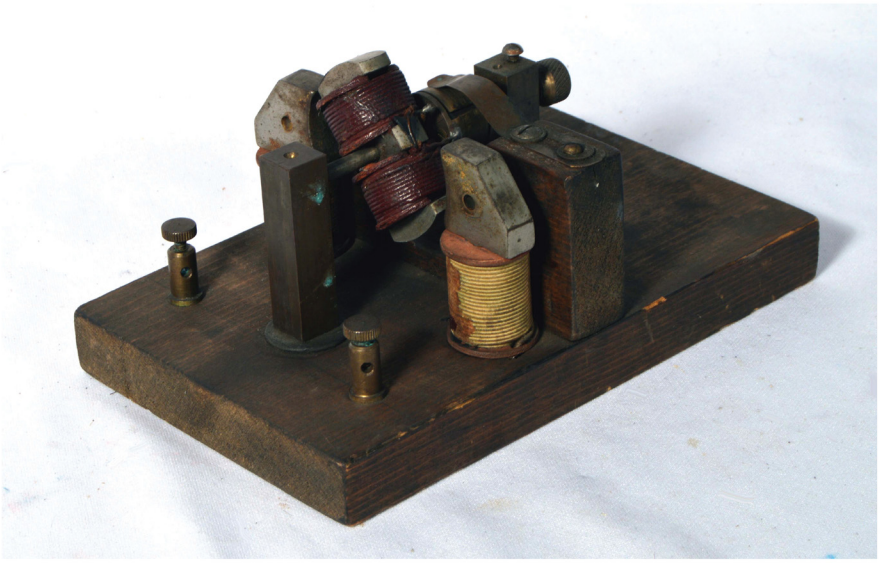
**121.** "Quantum biology," N. Lambert*et al.*, Nat. Phys. **9**(1), 10–18 (2013). Reviews the evidence for quantum mechanics playing a role in several biological systems, including the avian compass. (I)

**122.** "A model for photoreceptor-based magnetoreception in birds," T. Ritz, S. Adem, and K. Schulten, Biophys. J. **78**(2), 707–718 (2000). Presents a theoretical model for how the radical pair mechanism may underpin the avian compass and proposes cryptochrome as the specific molecule involved in the reaction. (A)

**123.** "Sustained quantum coherence and entanglement in the avian compass," E. M. Gauger, E. Rieper, J. L. Morton, S. C. Benjamin, and V. Vedral, Phys. Rev. Lett. **106**(4), 040503 (2011). Calculates that based on experimental data, if the radical pair theory is correct, the decoherence time of the involved electron spins is at least tens of microseconds—a surprisingly long time within a biological system. (A)

**124.** "Resonance effects indicate a radical-pair mechanism for avian magnetic compass," T. Ritz, P. Thalau, J. B. Phillips, R. Wiltschko, and W. Wiltschko, Nature **429**(6988), 177–180 (2004). Demonstrates that a weak magnetic field oscillating at 7 MHz may disrupt the European robin's magnetic compass, lending credibility to the radical pair theory. (A)

**125.** "Anthropogenic electromagnetic noise disrupts magnetic compass orientation in a migratory bird," S. Engels *et al.*, Nature **509**(7500), 353–356 (2014). Describes how ambient urban electromagnetic noise is sufficient to disrupt the robin's magnetic compass. (A)

**Small Direct-Current Motor**

This motor was abandoned in the back room of the Kenyon College lecture hall when I arrived in 1964. Somehow it survived, and is now in the Greenslade Collection. It appears to be locally-built, and was either made by Professor Elbe Johnson (who taught physics at Kenyon from 1911 to 1955) or by one of his students as a project. (Picture and Notes by Thomas B. Greenslade, Jr., Kenyon College)